NGUYEN THANH LUAN

0925752160 | ngthanhluan2511@gmail.com | github.com/Tluan-source

Summary

AI Engineer Intern candidate with a strong Computer Science foundation and experience in Computer Vision, NLP, RAG, and model fine-tuning. Seeking a full-time internship to build impactful AI applications and grow as an AI engineer.

Education

Ho Chi Minh City University of Technology and Engineering — Ho Chi Minh City, VN
Information Technology Engineer — *Aug. 2023 – Present*

- GPA: 3.2/4.0
- Relevant Coursework:Machine Learning, Reinforcement Learning, Natural Language Processing, Digital Image Processing, Mathematics for AI, Database Systems, Linear Algebra, Data Science

Projects

Face Anti-Spoofing System using CelebA-Spoof | *PyTorch, OpenCV, MobileNetV2, Computer Vision* — 2026
GitHub: github.com/Tluan-source/FSFM$_{L}$ite$_{P}$roject | Demo: huggingface.co/spaces/luan1232/face-anti-spoofing

- Developed a deep learning-based Face Anti-Spoofing system to distinguish genuine faces from presentation attacks using the CelebA-Spoof dataset.
- Built an end-to-end data pipeline including metadata processing, bounding-box based face cropping, context-aware facial region extraction, and image enhancement using CLAHE.
- Fine-tuned MobileNetV2 for binary classification (Live vs. Spoof) with data augmentation, transfer learning, and optimized training strategies.
- Evaluated model performance using Accuracy, ROC-AUC, Confusion Matrix, APCER, BPCER, and ACER metrics following Face Anti-Spoofing research protocols.
- Implemented real-time webcam inference pipeline for practical deployment and testing under varying lighting conditions.
- Managed large-scale dataset preprocessing and experimentation on Kaggle and local environments using Python and PyTorch.

Emotion-Aware RAG Platform | *Python, PyTorch, Transformers, NLP, Data Pipeline* — 2026
GitHub: https://github.com/Tluan-source/AI$_{E}$motion$_{R}$AG

- Conceptualized and initiated an Emotion-Aware RAG system designed to provide empathetic psychological assistance by coupling emotion detection with knowledge retrieval.
- Fine-tuned a Transformer-based Emotion Classification model to accurately detect psychological states (e.g., sadness, anxiety) from raw user text.
- Engineered an automated Knowledge Ingestion pipeline to extract, clean, and standardize large-scale unstructured medical text from diverse sources into structured formats.
- Implemented robust data preprocessing and validation strategies, including metadata tagging and text normalization, to prepare high-quality datasets for downstream VectorDB indexing.
- Established a modular architecture for data sourcing, ensuring scalable integration of future knowledge bases (e.g., PDFs, web crawlers, Hugging Face datasets).

Certifications

- TOEIC Listening & Reading: 635/990

Technical Skills

Programming Languages: Python, C/C++, SQL (SQLServer), JavaScript, Java, HTML/CSS
AI/ML: PyTorch, Scikit-learn, OpenCV, Computer Vision, Deep Learning, Machine Learning, NLP, CNNs, Data Augmentation, Model Evaluation
Data Science: Pandas, NumPy, Matplotlib, Seaborn, SPSS, Data Analysis, Statistical Analysis
Databases: MySQL, SQL Server
Tools and Platforms: Git, GitHub, Jupyter Notebook, VS Code, Linux, Hadoop, Apache Spark
Languages: Vietnamese (Native), English (Technical Reading Communication)